

Andrew Shi

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Summary

UC Irvine Computer Science undergraduate with experience building production full-stack systems, cloud applications, and LLM-powered agents. Seeking a 2027 Software Engineering internship.

Education

University of California, Irvine

Sep 2023 - Jan 2028

B.S., Computer Science, Minor in Mathematics | GPA: 3.8/4.0 | ICS Honors Program

Irvine, CA

• **Coursework:** Algorithms, Data Structures, Databases, Operating Systems, Computer Networks, Artificial Intelligence, Machine Learning

Experience

AitherPath

Jun 2026 - Sep 2026

Software Engineer Intern

Jersey City, NJ

- Built and shipped an internal conversational AI platform using **React**, **TypeScript**, **Cloudflare Workers**, and **Durable Objects**, adopted by 20+ teams, centralizing cross-application workflows while maintaining isolated, persistent state across real-time sessions.
- Built a unified agent execution layer integrating Gmail, Google Calendar, and Google Contacts across 15 **Zod-validated** tools, enabling employees to schedule events, look up contacts, and search, summarize, and extract information from their inbox through natural-language requests.
- Developed secure Google Workspace integration using **OAuth 2.0** authorization-code flows, implementing CSRF-protected state handling, scoped permissions, token exchange, and automatic access-token refresh to securely execute actions on employees' behalf.
- Engineered persistent conversational memory using **Cloudflare Agents** and **Vectorize**, combining token-budgeted context with semantic retrieval to preserve employee context and preferences across sessions, enabling employees to continue workflows without re-entering information.

Projects

FlixHub - Full-Stack Movie E-Commerce Platform <https://github.com/andrewshilw/flix-hub>

Jan 2026 - Mar 2026

- Built and deployed a full-stack movie platform using **Java Servlets**, **JDBC**, **MySQL**, and **REST APIs** on **AWS EC2/Tomcat**, implementing session-based authentication and checkout with HTTPS/TLS and parameterized queries.
- Engineered a CSV ETL pipeline that bulk-loaded ~**91K records in under 5 seconds** using batched JDBC inserts and in-memory keyset lookups; designed a MySQL stored procedure for atomic, deduplicated writes.
- Implemented full-text and fuzzy search using MySQL Boolean indexing and Levenshtein-distance matching, with debounced autocomplete and client-side caching for **sub-2-second** suggestions.
- Scaled the application horizontally by decomposing the monolith into login and movie services with **Redis-backed sessions**, MySQL replication, and JDBC connection pooling; containerized with **Docker** and deployed on **Kubernetes** with **load balancing** and Ingress routing.

PixelSphere - AI Integrated Social Network <https://github.com/andrewshilw/pixel-sphere>

May 2026 - Jun 2026

- Built a full-stack social networking platform using **React** and **Go**, implementing React Router v6 navigation, JWT-based stateless authentication, and OpenAI DALL·E 3 integration for on-demand AI image generation and editing.
- Developed containerized **Go microservices** for user and post workflows and deployed them on **Google App Engine** with Cloud Build-powered **CI/CD** and auto-scaling, maintaining **99%+ uptime**.
- Implemented full-text and filtered search with **Elasticsearch** deployed on **Google Compute Engine (GCE)**, enabling search across recent posts and user content with **sub-100ms query latency**.

FoodFlow - Online Food Ordering Web App <https://github.com/andrewshilw/food-flow>

Jun 2025 - Jul 2025

- Built a full-stack ordering platform using **Spring Boot**, **React**, and **Ant Design**, implementing **RESTful APIs** and responsive user workflows for registration, menu browsing, cart management, order placement, and checkout.
- Architected the Spring Boot backend using **object-oriented design** and a layered controller-service-repository architecture with dependency injection, implementing 15+ REST API endpoints for end-to-end order lifecycle management.
- Integrated **PostgreSQL** on **AWS RDS** using **Spring Data JDBC** and repository abstractions to manage relational user, restaurant, menu, and order data; implemented session-based authentication and authorization with **Spring Security**.
- Containerized the full-stack application with **Docker** and deployed it to **Google Cloud**, configuring automated builds and scalable deployment infrastructure while achieving **<200ms P99 API latency** under load testing.

Melodify - Android Music App <https://github.com/andrewshilw/melodify>

Mar 2026 - Apr 2026

- Built a **Kotlin** Android music application using **Android Jetpack**, architecting the app with **MVVM** and **Hilt** dependency injection for modular, testable, and maintainable code.
- Built a mock **REST API** with json-server and integrated **Retrofit** with asynchronous caching and request optimization, reducing API response times by **25%**; implemented **Room** persistence to support offline access and preserve user data across network interruptions.
- Developed feed, album, and favorites interfaces in **Jetpack Compose** with a persistent BottomBar and app-wide routing using **Jetpack Navigation**; implemented global audio playback with **Google ExoPlayer** for continuous playback across screens.

Technical Skills

- **Languages:** Java, Python, JavaScript/TypeScript, Go, SQL, C/C++, Kotlin, HTML/CSS
- **Frameworks:** Spring Boot, Spring Security, React, Next.js, Node.js, Express, Android Jetpack, Ant Design, Zod
- **Databases:** MySQL, PostgreSQL, Redis, Elasticsearch
- **Cloud & Infrastructure:** AWS, GCP, Cloudflare, Docker, Kubernetes, CI/CD
- **AI/LLM:** LangChain, RAG, Vector Embeddings, Vectorize, OpenAI API
- **Tools & Concepts:** Git, Maven, Gradle, Distributed Systems, Microservices, Object-Oriented Design